

Summary

Machine Learning Engineer with 5+ years of experience building scalable AI, machine learning, and analytics solutions across predictive modeling, NLP, and Generative AI applications. Skilled in Python, SQL, FastAPI, Flask, and cloud deployment on AWS with expertise in developing REST APIs, ETL pipelines, and RAG-based intelligent systems. Strong background in feature engineering, model optimization, statistical analysis, and data visualization. Experienced in MLOps practices, CI/CD workflows, Docker-based deployments, and scalable system architecture to deliver high-performance, data-driven business solutions.

Experience

Machine Learning Engineer | Devscape IT Solutions, USA

May 2024 – Feb 2026

- Developed scalable machine learning solutions on 120K+ records using XGBoost, KNN, Random Forest, and ensemble techniques for real-time prediction systems.
- Built high-performance REST APIs using Flask and FastAPI handling 10K+ monthly requests with optimized JSON processing and API validation through Postman.
- Improved model accuracy by 28% through hyperparameter tuning, cross-validation, feature engineering, and bias-variance optimization strategies.
- Increased pipeline efficiency by 34% by implementing automated preprocessing, feature selection, and scalable data transformation workflows.
- Designed Retrieval-Augmented Generation (RAG) systems using LLMs, FAISS, prompt engineering, and semantic search for intelligent document querying.
- Engineered NLP pipelines involving text preprocessing, embedding generation, tokenization, and contextual retrieval for AI-powered response systems.
- Deployed cloud-native ML applications on AWS and Render using Docker containers, REST architecture, and scalable backend microservices.
- Implemented CI/CD workflows using Git, GitHub, and automated deployment practices to improve development efficiency and release stability.
- Collaborated with cross-functional teams demonstrating strong communication, problem-solving, and leadership skills to deliver scalable AI solutions aligned with business goals.

Machine Learning Engineer | ZipNom Technologies, India

Jan 2021 – Sep 2023

- Increased loan prediction accuracy by 22% using classification algorithms including Logistic Regression, Random Forest, and ensemble learning models.
- Reduced misclassification rates by 18% through handling imbalanced datasets, feature selection, and improving overall model evaluation processes.
- Managed end-to-end machine learning workflows involving regression, clustering, and classification models for advanced financial risk analysis.
- Performed data cleaning, preprocessing, and exploratory data analysis using Python libraries including Pandas and NumPy to improve data quality.
- Designed optimized SQL queries using joins, CTEs, aggregations, and window functions for large-scale enterprise data processing workflows.
- Automated ETL pipelines for reliable ingestion, transformation, and validation of structured and unstructured enterprise business datasets.
- Improved MySQL database performance through indexing, schema optimization, query tuning, and advanced normalization strategy implementation techniques.
- Built interactive dashboards and visualizations using Power BI, Tableau, Matplotlib, and Seaborn for executive-level business performance reporting.
- Applied data structures, algorithms, and system design concepts to optimize application scalability and computational processing efficiency.

Skills

- Machine Learning & AI:** Regression | Classification | Clustering (K-Means, DBSCAN) | Random Forest | XGBoost | KNN | Ensemble Learning | NLP | Generative AI | LLMs | Prompt Engineering | RAG | Feature Engineering | Model Evaluation | Hyperparameter Tuning | Cross-Validation | PCA | Bias-Variance Optimization
- Programming & Data Analytics:** Python | SQL | Pandas | NumPy | Statistical Analysis | EDA | Hypothesis Testing | Correlation Analysis | Data Cleaning | Data Transformation
- Backend & APIs:** FastAPI | Flask | REST APIs | JSON | API Testing | Postman | Microservices Architecture
- Cloud, DevOps & MLOps:** AWS | Docker | CI/CD | Git | GitHub | Deployment Automation | ML Pipelines | Cloud Architecture | Model Deployment
- Databases & Data Engineering:** MySQL | ETL Pipelines | Query Optimization | Indexing | Schema Design | Data Warehousing Concepts
- Visualization & Tools:** Power BI | Tableau | Matplotlib | Seaborn | System Design | Data Structures & Algorithms

Education

Master of Science in Data Analytics

Dec 2025

Clark University, USA | GPA: 3.78

Bachelor of Technology in Electronics & Communication Engineering

May 2022

Mahaveer Institute of Science & Technology, India

Certifications

- AWS Certified Cloud Practitioner | Microsoft Power BI Data Analyst (PL-300) | Google Data Analytics Professional Certificate | Machine Learning Specialization – Coursera

Projects

Smart Document Query Assistant (RAG-Based AI System)

- Developed a Retrieval-Augmented Generation (RAG) system using LLMs, FAISS, and FastAPI for intelligent PDF-based query answering with high semantic retrieval accuracy.
- Designed scalable document ingestion, embedding generation, and prompt optimization pipelines deployed using Docker and AWS cloud environments.